Natural Hallucinations in Neural Machine Translation: A Comparative Review of Detection and Analysis Methods

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1 Introduction

Neural machine translation (NMT) is a state-of-the-art approach to machine translation that uses neural networks to produce more fluent and natural-sounding translations than ever before. Advances in NMT over the past decades have made these models increasingly accurate; still, these models have the potential to make errors, including hallucinations where the model's translations contain elements unrelated to the source input. While hallucinations in natural settings are rare, they can be detrimental when users of these systems take model correctness for granted (Oelschlager, 2024). Even the smallest chance that these translations contain hallucinations greatly compromise the extent to which users can trust these translations in real world applications.

Much research in NMT has focused on understanding, detecting, and mitigating hallucinations, but recent papers have noted this work is largely done in controlled settings with forced hallucinations. We specifically review Understanding and Detecting Hallucinations in Neural Machine Translation via Model Introspection by Xu et al. and Looking for a Needle in a Haystack by Guerreiro et al., both of whom pointed out this limitation, addressing the need to explore hallucinations in natural settings. In particular, Xu et al. study perturbed translations from both models translating English-Chinese and German-English to test hypotheses on symptoms of hallucinations. They use these results to inform tests on hallucination detection in natural settings. Guerreiro et al. solely focus on hallucination detection methods in natural settings, for which they use data from models translating German-English.

We begin by introducing a common framework for discussing these works. Then we outline motivations guiding this research. Finally, we evaluate the findings from these studies first by looking at symptoms of hallucinations and then at the detection of natural hallucinations.

2 Background

2.1 Definitions

While research surroundin-g hallucinations in LLMs has increased dramatically over the past few years, researchers consistently note that there is a lack of standardized methods for defining and classifying hallucinations (Guerreiro et al., 2023; Venkit et al., 2024; Xu et al., 2023). Moreover, the broad attributes that are ascribed to hallucinations under different frameworks can vary widely between sources and Natural Language Processing (NLP) tasks. A 2024 audit of hallucination frameworks in NLP research pointed to 8 distinct attributes commonly noted in definitions of hallucinations: fluency, plausibility, confidence, extrinsic, intrinsic, non-factual, unfaithfulness, and nonsensical (Venkit et al., 2024). They note papers in NMT have largely focused on extrinsic hallucinations – translations that contain information not present in the source content (Venkit et al., 2024). Indeed, both Guerreiro et al. (2023) and Xu et al. (2023) define hallucinations under what would be considered extrinsic hallucinations in the broad field of NLP and are careful to distinguish hallucinations from general translation errors. Additionally, both papers characterize NMT hallucinations by their unfaithfulness and both later note that

certain types of hallucinations also carry the nonsensical attribute.
Broadly, Xu et al. define hallucinations as translations that are “unfaithful to the input” (2023). They focus on what they call “detached hallucinations”, or hallucinations “where a major part of the output is unfaithful to the input” (Xu et al., 2023). These hallucinations are further categorized as “degenerated” or “non-degenerated”, with degenerated hallucinations being when the output contains n-grams that are repeated at least three more times than the original source sequence (Xu et al., 2023).
Guerreiro et al. likewise classify hallucinations in the context of the relationship between sources and translations. However, they explicitly note detachment in their definition of hallucinations, considering this foundational in discriminating between hallucinations and general errors (Guerreiro et al., 2023). They class these as either oscillatory or largely fluent, with these definitions aligning with those of degenerated and non-degenerated hallucinations, respectively. However, unlike Xu et al., Guerreiro et al. do not provide a strict definition for the number of repetitions necessary for a hallucination to be counted as oscillatory. Guerreiro et al. additionally evaluate the severity of largely fluent hallucinations, classing them as either fully detached or strongly detached. Their definitions for fully and strongly detached both fit under the larger framework of detached hallucinations outlaid by Xu et al.
We follow the general frameworks and definitions used by Xu et al. and Guerrerio et al. when referencing hallucinations:

- Hallucinations: translations where most or all of the output is unfaithful/detached from the source.
- Oscillatory hallucinations: hallucinations that contain repetitive n-grams in a way that is inconsistent with the source. We define degenerate hallucinations under oscillatory hallucinations.
- Largely fluent hallucinations: hallucinations that follow natural language patterns and are largely comprehensible. We consider non-degenerate hallucinations synonymous with largely fluent hallucinations.
- Translation errors: errors in translation that do not show detachment from the source.
- Reference translations: human-generated translations used as benchmarks for evaluating the quality of an automated translation.

Additionally, we talk about hallucinations in two ways as defined by Raunak et al.:

- Under Perturbation: hallucinations that are induced by introducing noise at the source.
- Natural: uninduced hallucinations.

2.2 Limitations of Past Research

Because natural hallucinations are quite rare in NMT, prior works have largely explored detection and classification methods using hallucinations generated in more “artificial” settings.

Source Perturbation Past studies have been conducted using perturbed data to generate translation data with plenty of hallucinations. Raunak et al. describe how source perturbations lead to a model’s amplification of the rarer training samples in the distribution rather than generalizing from the majority of the data (2021). Thus, these translations are often reproducing cases that are rare enough to likely not appear in natural translations; not only is perturbed data creating a disproportionate number of hallucinations, but it is also creating hallucinations that are not representative of ones that might be found in natural settings. The same study describes links between natural hallucinations and “corpus-level noise” (Raunak et al., 2021). Since data collected from the web in large scales will inevitably contain errors and invalid translations or misaligned text pairs, natural hallucinations will still occur in translations from NMT models trained on this type of data, only at a much lower rate. Though rare, the inevitable presence of hallucinations in translations emphasizes the importance of testing hallucination detection methods on natural settings to ensure these methods can be applied practically, even when real

cases are few and far between. Guerreiro et al. even liken hallucination detection in NMT to "finding a needle in a haystack."

Domain Shift In addition to source perturbations, past works have also largely operated under a domain shift, in which the model is trained on data from a specific domain, such as news articles, but faces difficulty in translating material from a different domain, such as technical manuals. Models tend to generate more hallucinations under domain shift compared to when they are trained with data that is more relevant to the material they will be translating. Both papers have a much stronger focus on source perturbation compared to domain shift.

3 Understanding Natural Hallucinations

3.1 Symptoms of Hallucinations

Xu et al. evaluate multiple hypotheses on how hallucinations are produced in NMT, focusing on various patterns in source token contributions (2023). The authors refer to these patterns as "symptoms" of hallucinations. Each hypothesis is tested by comparing hallucinations under perturbation to source text and reference-translations. Xu et al. argue that their results under perturbation can be generalized to natural hallucinations (2023), which we will discuss further when we consider their Layerwise Relevance Propagation-based classifier.
In this section, we will review the three hypotheses examined by Xu et al.

Low Source Contribution Hypothesis The Low Source Contribution Hypothesis predicts that hallucinations occur when an NMT model is too reliant on the target context over the source. This concept was first introduced by Voita et al. who analyze the relative source and target contributions in the model's generation process (2021). They found that models trained on more data are more likely to rely on source data for their translations and ultimately provide more reliable translations. They also found that models with exposure bias—in which the model relies on its own previous outputs to generate its next outputs without the ability to verify that previous outputs do not contain errors—tend to over-rely on the target and hallucinate more. Thus, they found a link between reliance on the target and hallucination generation. However, Voita et al.'s study only tested this hypothesis on pseudo-hallucinations that were not generated by NMT models.
Xu et al. finds that hallucinations under source perturbations have only slightly higher source contributions than the original samples as the contribution rate varied by (at most) 0.1 (Xu et al., 2023). This result conflicts with prior beliefs that pseudo-hallucinations had lower source contributions than reference translations; thus, the hypothesis does not necessarily hold across other types of hallucinations, so we cannot assume it is a reliable indicator of natural hallucinations.
Xu et al. explain this phenomenon by examining the contribution from the end-of-sequence (EOS) token in the source, as past works have also shown that hallucinations are likely to occur when the attention distribution is focused on this token. However, such findings were based only on qualitative analysis on individual samples. Xu et al. present quantitative results on the dataset of source-perturbed hallucinations that contradict this hypothesis, finding that the proportion of attention concentrated on the EOS token is not indicative of hallucinations, and the "proportion of source contribution from the EOS token is slightly higher on the original samples (11.2%) than that on the hallucinated samples (10.8%)" (Xu et al., 2023).

Local Source Contribution Hypothesis The Local Source Contribution Hypothesis proposes that "hallucinations occur when [an] NMT model overly relies on a small subset of source tokens across all generation steps" (Xu et al., 2023). This hypothesis was originally investigated in models of specific architectures with hallucinations caused by random token insertion, so Xu et al. aim to see if the hypothesis is generalizable to other architectures with other types of hallucinations.
The authors test this hypothesis by computing the High Contribution Ratio, positing if the Local Source Contribution Hypothesis holds true, this would provide a lower value on hallucinated examples than on

original samples (Xu et al., 2023). Indeed, this hypothesis held under this test, implying that hallucinations could be decreased by designing models that intentionally use a wider array of source tokens for generation.
Xu et al. further investigate whether the local source contribution has any bias towards the beginning, middle, or end of a source sequence. They find that hallucinated samples are linked to a disproportionately high source contribution at the beginning of a source sequence; contrastingly, the source contribution for original samples is higher at the end of the source sequence (Xu et al., 2023). This increase in contribution suggests that models use ends of source sequences as indicators for when generation should finish.

Static Source Contribution Hypothesis The Static Source Contribution Hypothesis states that, when a model hallucinates, the distribution of contributions from different source tokens remains static across all inference steps. Xu et al. create this original hypothesis as a formalization of similar postulations from prior works; however, unlike many past studies that focus on the "static attention" to EOS tokens, this hypothesis does not focus on any particular source token (Xu et al., 2023).
Xu et al. use both attention and layerwise relevance propagation (LRP), which measures relative influence scores of different input tokens at different layers of the model, to quantify the model's focus on certain tokens. By visualizing the source contributions at different positions of original and hallucinated samples, Xu et al. observed that the position of source contributions in original samples shifts dynamically among different target generation steps, but the source contribution distribution remains relatively constant across generation steps when the model hallucinates (Xu et al., 2023). From these findings, it is clear that the Static Source Contribution Hypothesis holds. Their results also reveal that LRP more effectively distinguishes hallucinations from non-hallucinations when compared with attention.
The staticity of hallucinating models makes sense: when the model focuses on the same components of the source across all inference steps, it likely ignores information that would create a more accurate translation.

3.2 Detecting Hallucinations

Both Xu et al. and Guerreiro et al. propose and analyze several methods for detecting natural hallucinations. Xu et al. examine ten detection methods on models translating English-Chinese and models translating German-English, whereas Guerriero et al. examine nine detection methods on models translating German-English. Both papers use hand-annotated datasets for testing the accuracy of such methods. It is important to note that the comparison of their results is caveated by differences in datasets and annotation frameworks.
In this section, we review five detection methods discussed by these papers.

COMET COMET (Rei et al., 2020) has been widely adopted in machine translation research as a framework for evaluating the quality of model outputs using human-generated, reference translations. While this method is considered "state-of-the-art" for model evaluation, the requirement of reference translations makes it unsuitable for deployment in on-the-fly applications. Guerreiro et al. instead use this method as a standard for understanding the performance of other detection measures, considering the performance of COMET as an upper bound (2023).
Indeed, COMET outperforms all other hallucination detection methods tested by Guerreiro et al., correctly flagging 75% of hallucinations in the test set. Of these, COMET was best at identifying fully-detached hallucinations (nearly 100%) but performs similarly well at identifying oscillatory hallucinations (over 80%), and decently at identifying strongly-detached hallucinations (around 50%) (Guerreiro et al., 2023). These results align with intuition – more severe abnormalities are easier to detect. Because oscillatory hallucinations are misaligned with common natural language patterns, it also makes sense that these would be viewed as lower quality, and, thus, more likely to be flagged by COMET.

COMET-QE COMET-QE is the quality estimator counterpart of COMET, meaning it does not rely on reference translations when predicting the quality of an output. Kocmi et al. (2021) and Freitag et al. (2021) both performed extensive evaluations of various quality evaluation and estimation techniques in

machine translation. They find COMET-QE performs on par with COMET under certain performance metrics, performing especially well at the system-level (Kocmi et al., 2021; Freitag et al., 2021), while Freitag et al. noted its results were variable at the segment-level (2021).
When evaluating COMET-QE as a hallucination detection method, Guerreiro et al. finds that it performs significantly worse than COMET, only correctly flagging 35% of hallucinations in the test set. In fact, this method performs in the bottom 50% of all methods tested, notably failing to detect nearly all fully-detached and strongly-detached hallucinations. Xu et al. also evaluates COMET-QE as a hallucination detector but contrastingly finds it operates with fairly high accuracy (around 70% for both an English-Chinese classifier and German-English classifier) (2023). However, it significantly over-flags correct translations produced by the English-Chinese classifier (32.4% precision) (Xu et al., 2023), supporting Guerreiro et al.'s skepticism toward using this model as a detection method. It is also worth noting that since Xu et al. did not annotate hallucinations in the exact manner as Guerreiro et al., it is difficult to directly compare these results.
The discrepancies in system- versus segment-level performance noted previously could explain why COMET-QE performs variably. Guerreiro et al. posit COMET-QE "fails to rank by the severity of a translation pathology" supporting that it has more trouble with certain segment-level tasks. Still, COMET-QE detects oscillatory hallucinations at nearly the same rate as COMET (Guerreiro et al., 2023). Again, the unique structure of these hallucinations may act as an indicator of lower translation quality to these models.

Attention Patterns As mentioned in the previous section, prior literature associates hallucinations with concentrated attention mass on the EOS token, taking this as an indicator that the model is ignoring the input (Guerreiro et al., 2023). Based on this, Guerreiro et al. suggest two detection heuristics: Attn-to-EOS, "the proportion of attention paid to the EOS source token" and Attn-ign-SRC, "the proportion of source words with a total incoming attention mass lower than 0.2" (2023). These heuristics produce contrasting results, with Attn-to-SRC being one of the top performing detectors, flagging 60% of all hallucinations in the test set. On the other hand, Attn-to-EOS was one of the lowest performing detectors, only flagging 13% of all hallucinations (Guerreiro et al., 2023). These results align with Xu et al.'s findings for the Low Source Contribution Hypothesis as they show the proportion of attention focused on the EOS token is not indicative of hallucinations. This suggests a shift in our understanding of hallucination indicators.

Layerwise Relevance Propagation Based on their evaluation of hallucination symptoms relative to source token contributions, Xu et al. implement an LRP-based classifier for detecting hallucinations (2023). This detector is posed as an alternative to attention-based classifiers, aligning with their findings on source contribution staticity. Ultimately, Xu et al. determine the LRP classifier to be the best overall detection method of the ones they tested, showing it achieved the highest F1 scores and identified 76.2% and 86.5% of all hallucinations in the German-English and English-Chinese test sets, respectively (2023). Importantly, they argue the success of this model indicates their results on hallucinations under perturbation are generalizable to natural hallucinations (Xu et al., 2023).

Measures of Confidence Guerreiro et al. note that uncertainty measures have successfully aided quality estimation in the past, hypothesizing that this can be extended to hallucination detection (2023). To test this, they apply sequence log-probability, a measure to calculate model confidence. Results of applying this method as a hallucination detector yielded great success, with sequence log-probability flagging 73% of hallucinations in the test set – just 2% less than COMET (Guerreiro et al. 2023). Sequence log-probability even outperforms COMET when detecting largely fluent hallucinations, though it falls short for oscillatory hallucinations (Guerreiro et al. 2023). From these results, Guerreiro et al. posit that model confidence acts as an indicator of hallucination likelihood (2023).
Xu et al. also test measures of model confidence as hallucination detectors through an NMT probability score. However, their results contradict Guerreiro et al.'s, showing poor performance nearly across the board when using this heuristic (Xu et al. 2023). In particular, this method had a recall of 3.4% when tested on a German-English classifier (Xu et al. 2023), which is strikingly low relative to Guerreiro et al.'s

results noted above. Once again, the variance in test sets between the two papers makes it difficult to draw direct comparisons. Still, such a large difference in results despite consistent frameworks suggest the cause is likely not a difference in test sets but rather how thresholds are set when tuning these metrics.

Ensemble Methods In attempts to increase model precision, Guerreiro et al. additionally test the usage of ensemble methods for detecting hallucinations by passing sequence log-probability flagged translations through COMET-QE (2023). However, they found this came at the expense of recall, with COMET-QE filtering out the vast majority of correct hallucinations flagged by sequence log-probability (Guerreiro et al., 2023). Xu et al. had very similar results when testing multiple ensemble classifiers, likewise noting the precision-recall tradeoffs of such methods (2023).

4 Conclusion and Future Directions

In this paper we review two recent studies on hallucinations in natural settings. We discuss the hypotheses evaluated by Xu et al. and the detection methods proposed by both Xu et al. and Guerreiro et al. to analyze natural hallucinations in NMT models. In a world growing increasingly dependent on language models, accurate hallucination detection is key for model reliability and user trust.

We acknowledge that the scope of this paper is limited strictly to extrinsic, detached hallucinations, so the expressed findings are not necessarily generalizable to other types of hallucinations in a non-NMT context. Additionally, though both studies we focused on used a German-English language pair, they annotated separately and had slight discrepancies in how they annotated. This may compromise the comparability of the two papers' findings.

4.1 Future Directions for Research

Since only two language pairs are used between Xu et al. and Guerreiro et al., following the same methodologies with different language pairs could help create a more robust understanding of how effective the proposed detection methods are.

The generalizability of these symptoms and detection methods across other non-NMT areas of NLP would also be useful to move towards language models with less misinformation; specifically, future studies could determine whether these detection methods can accurately classify intrinsic hallucinations as well as extrinsic ones.

Finally, we echo the call of many past researchers for a standardized framework for evaluating hallucinations. By systematizing the categorization of hallucinations, we can ensure that future findings can be reliably compared against one another to create a more robust understanding of hallucinations and how to properly prevent them.

References

Markus Freitag, Ricardo Rei, Nitika Mathur, Chi-kiu Lo, Craig Stewart, George Foster, Alon Lavie, and Ondřej Bojar. 2021. Results of the WMT21 Metrics Shared Task: Evaluating Metrics with Expert-based Human Evaluations on TED and News Domain. In *Proceedings of the Sixth Conference on Machine Translation*, pages 733–774, Online. Association for Computational Linguistics.

Nuno M. Guerreiro, Elena Voita, and André Martins. 2023. Looking for a Needle in a Haystack: A Comprehensive Study of Hallucinations in Neural Machine Translation. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1059–1075, Dubrovnik, Croatia. Association for Computational Linguistics.

Tom Kocmi, Christian Federmann, Roman Grundkiewicz, Marcin Junczys-Dowmunt, Hitokazu Matsushita, and Arul Menezes. 2021. To Ship or Not to Ship: An Extensive Evaluation of Automatic Metrics for Machine Translation. In *Proceedings of the Sixth Conference on Machine Translation*, pages 478–494, Online. Association for Computational Linguistics.

Pranav Narayanan Venkit, Tatiana Chakravorti, Vipul Gupta, Heidi Biggs, Mukund Srinath, Koustava Goswami, Sarah Rajtmajer, and Shomir Wilson. 2024. An Audit on the Perspectives and Challenges of Hallucinations in NLP. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6528–6548, Miami, Florida, USA. Association for Computational Linguistics.

Richard Oelschlager. 2024. Evaluating the Impact of Hallucinations on User Trust and Satisfaction in LLM-based Systems. Dissertation.

Vikas Raunak, Arul Menezes, and Marcin Junczys-Dowmunt. 2021. The Curious Case of Hallucinations in Neural Machine Translation. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1172–1183, Online. Association for Computational Linguistics.

Ricardo Rei, Craig Stewart, Ana C Farinha, and Alon Lavie. 2020. COMET: A Neural Framework for MT Evaluation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702, Online. Association for Computational Linguistics.

Elena Voita, Rico Sennrich, and Ivan Titov. 2021. Analyzing the Source and Target Contributions to Predictions in Neural Machine Translation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1126–1140, Online. Association for Computational Linguistics.

Weijia Xu, Sweta Agrawal, Eleftheria Briakou, Marianna J. Martindale, and Marine Carpuat. 2023. Understanding and Detecting Hallucinations in Neural Machine Translation via Model Introspection. *Transactions of the Association for Computational Linguistics*, 11:546–564.